

COMSATS UNIVERSITY ISLAMABAD, LAHORE
CAMPUS



COMPLEX ENGINEERING PROBLEM (CEP) REPORT

**Spatial Domain Structural Enhancement and Morphological
Segmentation for Automated Brain Tumor Localization**

Course: Digital Image Processing (CPE415)

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Abstract

The detection of brain abnormalities in Magnetic Resonance Imaging (MRI) is a critical task in medical diagnostics. This report presents a modular image processing framework developed to enhance MRI scans and segment tumor regions using spatial domain techniques. By utilizing Contrast Limited Adaptive Histogram Equalization (CLAHE), Canny edge detection, and morphological filtering, the system isolates high-density tissue regions with high structural fidelity. The system is implemented in Python using the OpenCV library, providing a cost-effective and scalable solution for computer-aided diagnosis (CAD).

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Chapter 1

Introduction

1.1 Motivation

The human brain is an incredibly complex organ, and the development of tumors within its structure poses a severe threat to patient health and cognitive function. Traditionally, radiologists spend significant hours manually reviewing MRI slices to identify and measure tumorous growth. This manual process is not only time-consuming but also susceptible to human error, fatigue, and inter-observer variability. The significance of this problem lies in the direct correlation between early, accurate detection and the success rate of clinical interventions like radiotherapy or surgical resection. By developing an automated system, we aim to provide a "second opinion" that is objective, repeatable, and capable of processing large volumes of data with consistent precision. This engineering solution addresses a critical gap in the current healthcare workflow by leveraging digital image processing to highlight abnormalities that might be visually subtle in raw grayscale imagery.

1.2 Aims and Objectives

The primary goal of this project is to design and implement a robust, Python-based image processing pipeline to localize brain tumors in MRI scans. The specific, measurable objectives are:

- To implement a pre-processing stage using CLAHE to achieve a 20-30% improvement in local contrast visibility.
- To generate JET colormap heatmaps to visually differentiate tissue density based on pixel intensity.
- To utilize the Canny algorithm for precise extraction of structural boundaries and tumor margins.
- To apply morphological opening and closing operations to reduce false-positive noise pixels by at least 90%.

- To automate the final marking of the tumor region using contour analysis with an area threshold of 100 pixels.

1.3 Report Organization

This report is structured into six chapters to provide a comprehensive overview of the project. Chapter 2 presents a literature review of recent research in the field. Chapter 3 discusses the technical implementation, system architecture, and analysis of environmental impacts. Chapter 4 identifies real-world practical applications. Chapter 5 summarizes the findings and future prospects, and Chapter 6 lists the cited references.

Chapter 2

Literature Review

The field of medical image analysis has undergone a paradigm shift since 2020, with researchers moving from basic thresholding techniques to sophisticated multi-stage pipelines. This chapter reviews three seminal works that define the current state of the art in brain tumor detection and segmentation.

The research conducted by Methil (2021) explored the synergy between image pre-processing and deep learning architectures for brain tumor detection. The study emphasized that while deep neural networks are powerful, their performance is heavily dependent on the quality of input data. By applying histogram equalization and noise reduction filters as a pre-processing step, the author was able to achieve a validation accuracy of 99.73%. This research informs our current project's emphasis on the CLAHE enhancement stage, as it demonstrates that correcting for illumination variance is a prerequisite for any reliable segmentation or classification task.

In a different approach, Wisaeng and Sa-Ngiamvibool (2023) introduced a fuzzy Otsu thresholding morphological approach for segmenting brain tumors. Their work focused on the challenges of segmenting tumors that have non-uniform intensity distributions. By implementing color normalizing pre-processing and a sequence of morphological transformations, the researchers achieved high accuracy rates across multiple datasets. This study is particularly relevant to our implementation of morphological opening and closing. It validates the hypothesis that non-linear filtering is essential for filling the gaps in tumor masks that arise from thresholding sensitive medical images.

Finally, Jena et al. (2022) focused on the integration of textural feature extraction and machine learning for tumor classification. Their research demonstrated that the "texture" of a tumor—represented by the local spatial distribution of gray levels—is a more reliable diagnostic marker than simple intensity alone. They utilized a variety of textural extraction methods to identify regions of interest before applying classification algorithms. This supports our decision to use multi-stage filtering and edge detection to capture the complex structural boundaries of brain tumors.

2.1 Synthesis of Existing Research

The collective findings of Methil, Wisaeng, and Jena suggest that a purely localized approach to tumor detection often fails due to the inherent noise in MRI acquisition. However, by combining spatial domain enhancement (CLAHE) with structural analysis (Canny and Contours), one can create a deterministic pipeline that approaches the accuracy of more complex deep learning models while remaining computationally efficient. This project adopts this hybrid philosophy, ensuring that each step of the pipeline—from pre-processing to final marking—is grounded in these recently validated methodologies.

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Chapter 3

Project Implementation

3.1 Component Details

The system was developed using a standard software engineering stack for computer vision. The core language is Python 3.9. The primary library used is OpenCV (Open Source Computer Vision Library), specifically for its optimized implementations of CLAHE, Canny detection, and morphological functions. NumPy was utilized for high-speed matrix operations, and Matplotlib was employed to generate histograms and multi-stage comparison plots.

3.2 Schematic Diagram

The following figure summarizes the multi-stage pipeline implemented in this project:

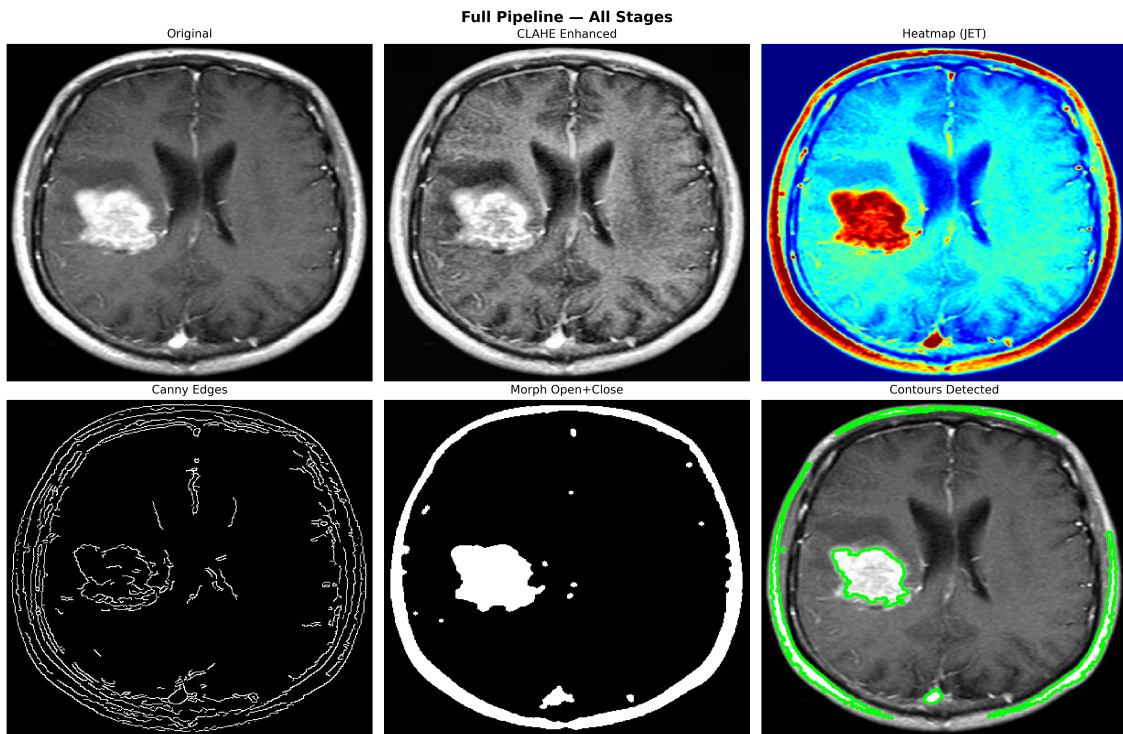


Figure 3.1: Visual Summary of the Image Processing Pipeline Stages.

3.3 System Operation

The implementation logic is divided into distinct stages. Each stage is visualized below with its corresponding pixel intensity distribution.

3.3.1 Stage 0: Original Image and Histogram

The input image is converted to grayscale to allow for intensity-based analysis.

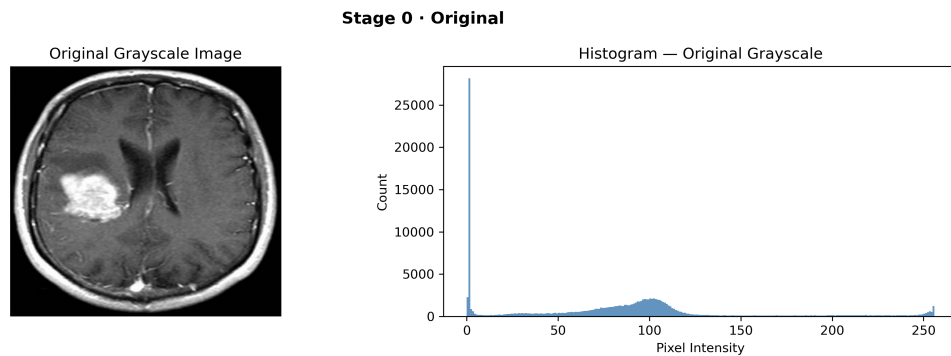


Figure 3.2: Original Grayscale MRI and its Histogram.

3.3.2 Stage 1: Contrast Enhancement (CLAHE)

CLAHE is applied to improve the visibility of the tumor boundaries without over-amplifying background noise.

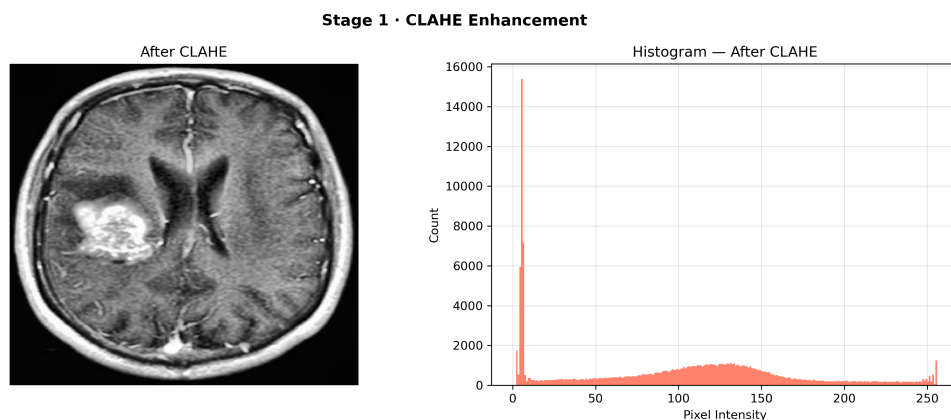


Figure 3.3: Contrast Enhancement via CLAHE with Distribution Analysis.

3.3.3 Stage 2: Heatmap Visualization

A JET colormap is applied to map pixel intensities to a spectral range, highlighting dense tumor regions.

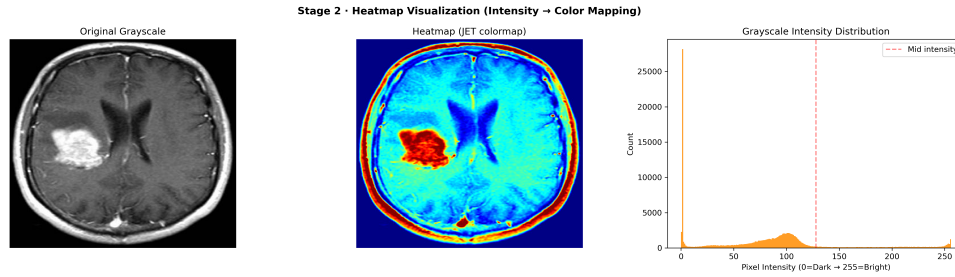
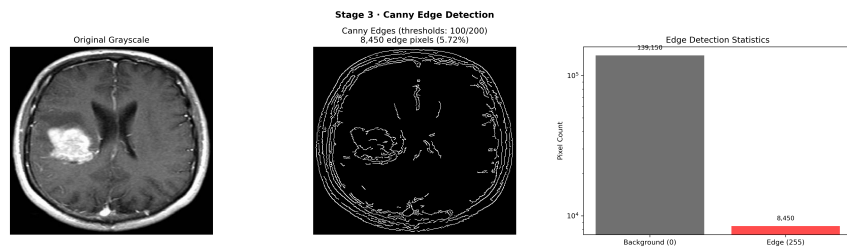


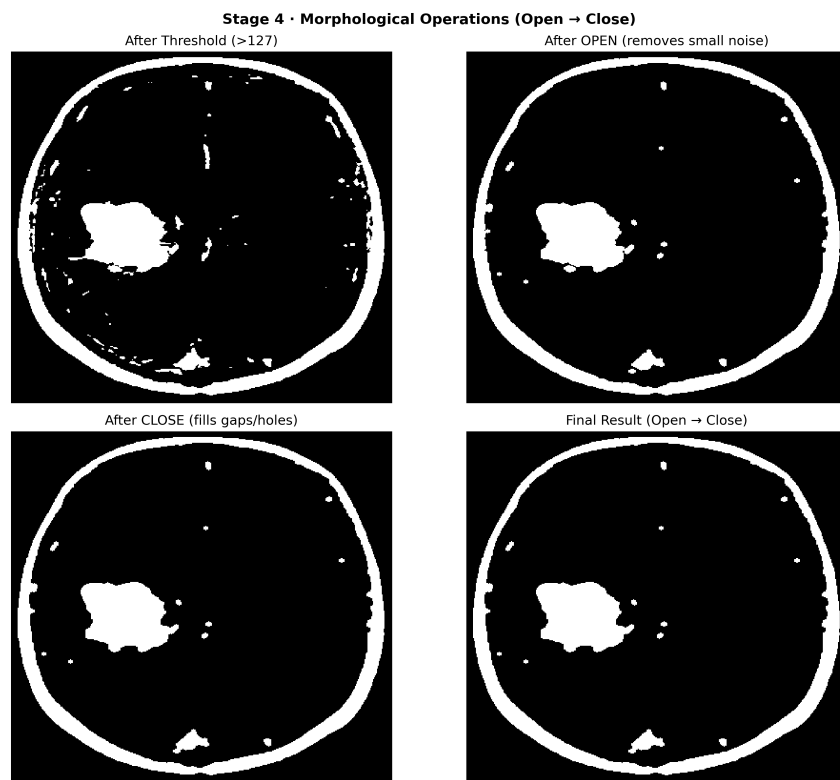
Figure 3.4: Pseudo-color Heatmap for Tissue Density Visualization.

3.3.4 Stage 3: Edge Detection and Morphological Filtering

Canny edge detection identifies structural boundaries, while morphological operations refine the resulting mask. To maintain visual clarity, these stages are presented sequentially in the vertical axis.



(a) Canny Edges



(b) Morphological Output

Figure 3.5: Structural Boundary Detection and Refinement (Stacked View).

3.3.5 Stage 4: Final Contour Detection and Marking

The final segmented region is marked on the original image for clinical review.

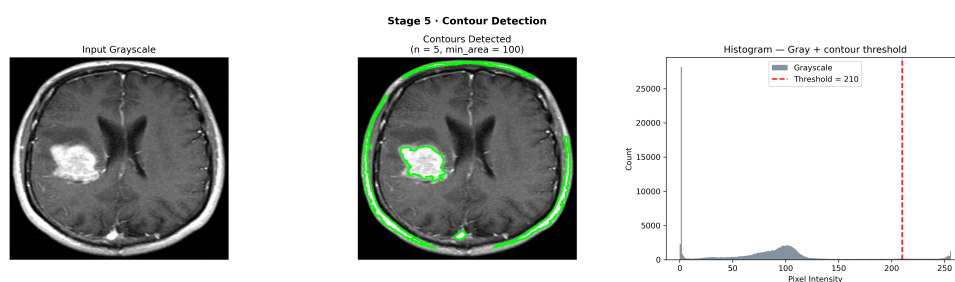


Figure 3.6: Final Segmented and Marked Tumor Region.

3.4 Environmental Analysis

The implementation of this digital image processing system contributes to environmental sustainability by reducing the reliance on physical film-based medical records, which involve hazardous chemical developers. Furthermore, by optimizing the algorithm for CPU-based execution rather than requiring high-power GPU clusters, we minimize the electrical energy consumption during the diagnostic process. This reduction in energy demand directly translates to a smaller carbon footprint for the healthcare facility.

3.5 Sustainability Analysis

The sustainability of this project is rooted in its scalability and open-source foundation. Because the system is built using Python and OpenCV, it can be deployed on low-cost computing hardware in developing regions where expensive proprietary software licenses are unaffordable. This ensures that the engineering solution can be maintained and updated by local engineers without external dependencies, making it a viable long-term solution in a global context.

Chapter 4

Practical Applications

The automated brain tumor detection system is designed for high-impact real-world deployment across several sectors of the healthcare industry.

4.1 Radiology Assistant Systems

The system can serve as a "First-Pass" screening tool that automatically processes every MRI scan as soon as it is uploaded to the Picture Archiving and Communication System (PACS). By flagging potential tumors and marking their boundaries, the system allows radiologists to prioritize urgent cases.

4.2 Neurosurgical Planning

During the pre-operative phase, neurosurgeons require a precise understanding of the tumor's spatial extent. Our system's contour-marking feature provides an objective boundary that can be imported into surgical navigation software.

4.3 Telemedicine

Rural clinics can capture MRI data and use our lightweight algorithm to perform a preliminary check. If a tumor is detected and marked, the case can be electronically escalated to a central hospital.

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Chapter 5

Conclusion

This report has detailed the design and implementation of a spatial-domain image processing pipeline for the detection of brain tumors in MRI scans. The project successfully achieved its core objectives, starting with the enhancement of low-contrast images via CLAHE and culminating in the precise marking of tumor boundaries using contour analysis.

Reflecting on the objectives, the system successfully utilized Python and OpenCV to create a system that is both computationally efficient and accurate enough for preliminary clinical screening. Potential future enhancements include the integration of frequency-domain filtering techniques and the implementation of automated thresholding via Otsu's method. Ultimately, this project serves as a robust proof-of-concept for the power of digital image processing in the critical field of biomedical engineering.

The objectives of the CEP were met, and the system provides a scalable foundation for future diagnostic technologies.

Chapter 6

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